GEOMETRY - HANDOUT 01

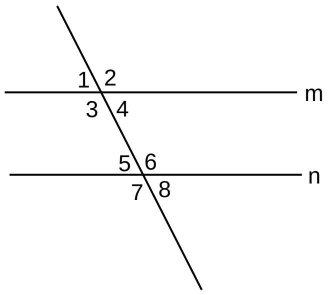
Directions for questions 1 to 20: Solve the following questions.

1. (i) Find the supplement of an angle whose complement is three and half times of its measure.

(ii) The complementary and the supplementary angles of an angle are in the ratio 1 : 3. Find the angle.

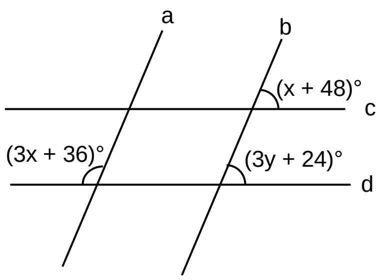
2. Lines L_1 and L_2 meet at point P. The four angles that are formed, in an order, are A, B, C and D. What will be the measure and the type of the angles A, B, and C if angle D is
(i) 85°? (ii) 105°? (iii) acute?

3. (i)



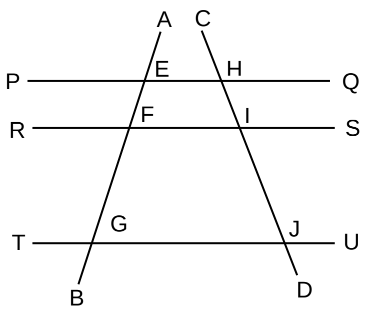
In the figure, lines m and n are parallel. Seven times the measure of angle 2 is equal to eleven times the measure of angle 4. What is the measure of angle 7?

(ii)



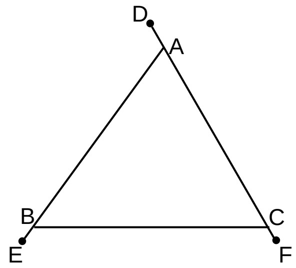
In the figure, lines a and b are parallel. Lines c and d are also parallel. What is the value of y?

4. In the figure, PQ || RS || TU. AB and CD are the transversals cutting these parallel lines as shown.



5. (i) In isosceles triangle ABC, the external angle at C measures 130°. The largest possible measure of any of the angles of the triangle is ______.

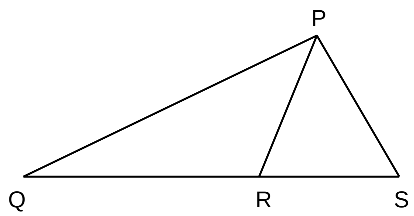
(ii)



In the figure, $\angle$DAB + $\angle$EBC = 250° and $\angle$EBC + $\angle$BCF = 230°. Find the measure of $\angle$ABC.

6. In a triangle, the measures of all the angles (in degrees) are integers. One of the angles of the triangle is 50°. How many combinations of the angles of the triangle are possible if the triangle is
(i) acute?
(ii) scalene?

7.



In the figure, $\angle$PSQ = 75° and triangle PQR is isosceles. If $\angle$PQR = 30°, find the measure of $\angle$RPS.

8. The lengths of the sides of four triangles are given below. Identify the type of triangle in each case.
(i) 7, 24, 25
(ii) 9, 10, 18
(iii) 8, 8, $8\sqrt{2}$
(iv) 8, 10, 18

9. A scalene triangle has integral sides and a perimeter of 30. How many such triangles are possible?

10. (i) G is the centroid of triangle ABC. AB = 48 m, BC = 14 m and AC = 50 m. BD is the median drawn to AC. Find the measure of GD.

(ii) G is the centroid of triangle PQR. If the area of PQR is 30 sq units, the area of triangle GQR is ______.

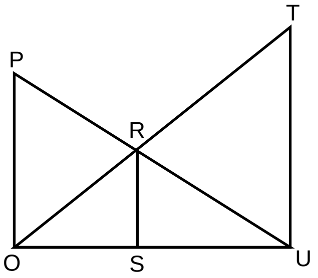
11. (i) The base of an isosceles triangle is 28 m long. The altitude drawn to the base is $2\sqrt{15}$ m long. Find the perimeter of the triangle.

(ii) In triangle XYZ, XY = 10 m, YZ = 24 m and XZ = 26 m. The length of the altitude to XZ is ______.

12. (i) The bisectors of ∠B and ∠C in triangle ABC meet at I. If ∠BAC = 48°, find ∠BIC.

(ii) In triangle PQR, the internal angle bisector of angle P meets QR at S. If PQ = 15 m, PR = 25 m and QR = 32 m, find QS.

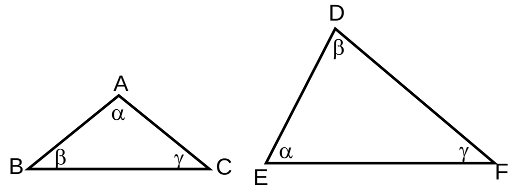
13.



In the figure, ∠PQU = ∠TUQ = ∠RSU = 90°. If PQ = 36, TU = 45 and SU = 29.7, find RS.

14. (i) In ΔPQR, points A and B lie on the sides PQ and PR respectively, such that 3PA = 4AQ and 3PB = 4BR. If QR = 70 cm, find AB.

(ii) In the figure, ABC and DEF are triangles such that AC = 32 cm, BC = 40 cm and DE = 60 cm. Find the perimeter of triangle ABC, if the perimeter of triangle DEF is 186 cm.



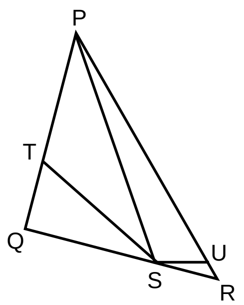
15. In quadrilateral ABCD, ∠ABD = ∠DBC and ∠ADB = ∠BDC. Find the ratio of the areas of triangles ABD and BCD.

ADDITIONAL QUESTIONS FOR PRACTICE

Directions for questions 1 to 5: Solve the following questions.

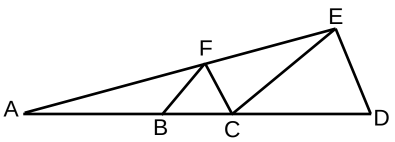
1. Two sides of a triangle are 8 cm and 15 cm. The measure of the third side (in cm) is an integer. If the triangle is acute, how many possible values are there for the measure of the third side?

2.



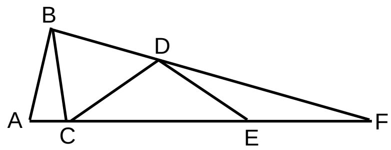
In the figure, PQR is a triangle and S is a point on QR such that QS : SR = 2 : 1. The bisectors of angles PSQ and PSR meet PQ and PR at T and U respectively. If PU = 128 cm, UR = 16 cm and PQ = 100 cm, find PT.

3.



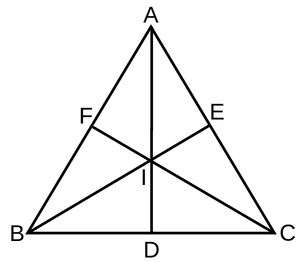
In the figure, BF || CE and CF || DE. If BC = 2 cm and CD = 3 cm, find AB.

4.



In the figure (not to scale), AB = BC = CD = DE = EF. If ∠ABC = ∠DFE, find ∠DFE.

5.



In the figure, AD, BE and CF are the bisectors of ∠BAC, ∠ABC and ∠ACB respectively. If AC = BC and AB = 2BD, find ∠BIC.